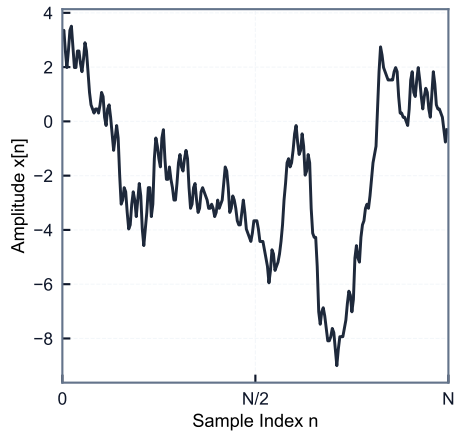
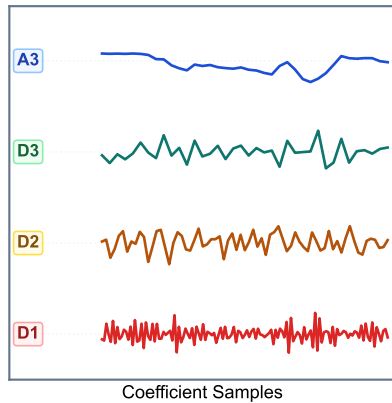


(a) Raw EEG Sequence $x[n]$



(b) Wavelet Sub-Bands (A_3, D_3, D_2, D_1)



(c) 6 Non-Linear Descriptors

1. Energy: $E_j = \sum |c_{j,k}|^2$
2. Std Dev: $\sigma_j = \text{std}(c_j)$
3. Skewness: $S_j = \mathbb{E}[(c - \mu)^3] / \sigma^3$
4. Kurtosis: $K_j = \mathbb{E}[(c - \mu)^4] / \sigma^4$
5. Wiener Ent: $H_j = \frac{\exp(\mathbb{E}[\ln c^2])}{\mathbb{E}[c^2]}$
6. PSD RMS: $\text{RMS}_j = \sqrt{\frac{1}{M} \sum P_j(f)}$

(d) Wavelet Tensor $\mathbf{x}_{\text{DWT}}$

1. Sub-Band Vector:
 $\mathbf{d}_j = [E_j, \sigma_j, S_j, K_j, H_j, \text{RMS}_j]^\top$
2. Channel Vector:
 $\mathbf{d}_c = [\mathbf{d}_{A3}^\top, \mathbf{d}_{D3}^\top, \mathbf{d}_{D2}^\top, \mathbf{d}_{D1}^\top]^\top$
3. Channel Dim:
 $\dim(\mathbf{d}_c) = 4 \times 6 = 24\text{D}$
4. Multichannel:
 $\mathbf{x}_{\text{DWT}} = [\mathbf{d}_1^\top, \dots, \mathbf{d}_C^\top]^\top$

$$\mathbf{x}_{\text{DWT}} \in \mathbb{R}^{C \times 24}$$

($C = 18 \rightarrow 432\text{D}$)